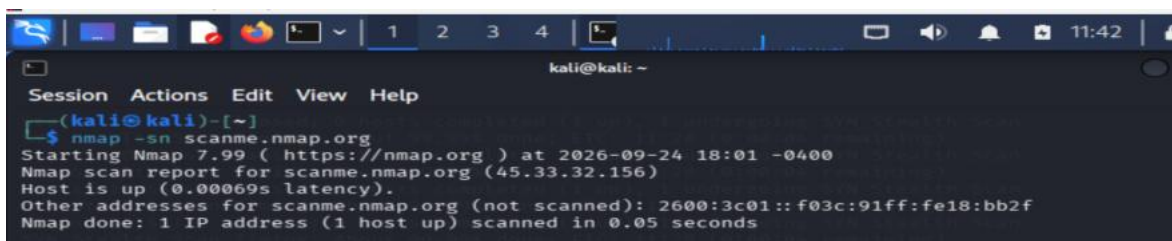


Stephania de Jesús Rivera Morales.

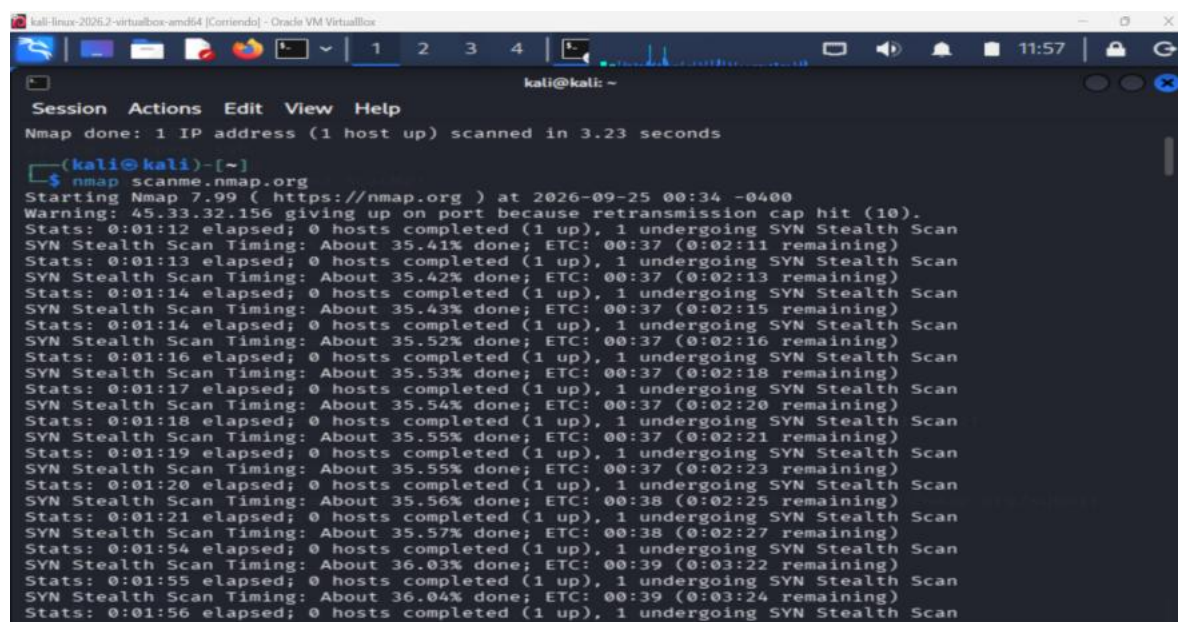
Hacker Women Council.



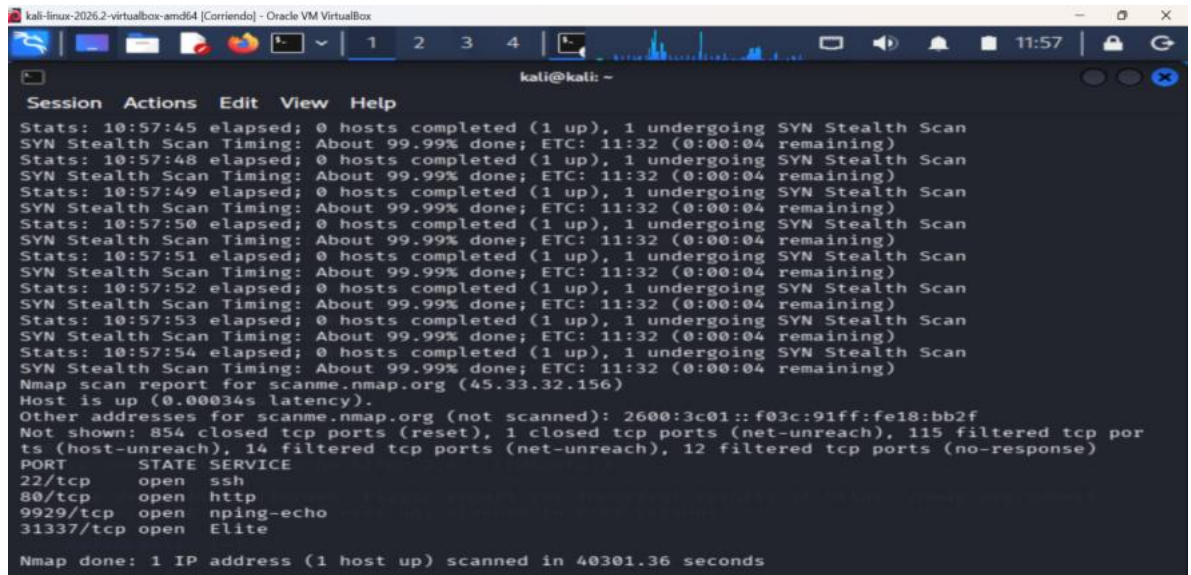
```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)~  
$ nmap -sn scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 18:01 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.00069s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
```

1. Verifica el objetivo

- **Comando utilizado:** `nmap -sn scanme.nmap.org`
- **Evidencia y Resultado:** El comando confirma que el host remoto scanme.nmap.org (IP 45.33.32.156) está activo y responde correctamente a las solicitudes de sondeo de red (Ping Scan) con una latencia de 0.00069s.
- **Observaciones:** A diferencia de la entrega anterior realizada en el entorno local (127.0.0.1), este comando valida la alcanzabilidad de un objetivo externo real en Internet, cumpliendo con el alcance del Challenge.



```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 3.23 seconds  
(kali@kali)~  
$ nmap scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 00:34 -0400  
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).  
Stats: 0:01:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.41% done; ETC: 00:37 (0:02:11 remaining)  
Stats: 0:01:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.42% done; ETC: 00:37 (0:02:13 remaining)  
Stats: 0:01:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.43% done; ETC: 00:37 (0:02:15 remaining)  
Stats: 0:01:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.52% done; ETC: 00:37 (0:02:16 remaining)  
Stats: 0:01:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.53% done; ETC: 00:37 (0:02:18 remaining)  
Stats: 0:01:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.54% done; ETC: 00:37 (0:02:20 remaining)  
Stats: 0:01:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.55% done; ETC: 00:37 (0:02:21 remaining)  
Stats: 0:01:19 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.55% done; ETC: 00:37 (0:02:23 remaining)  
Stats: 0:01:20 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.56% done; ETC: 00:38 (0:02:25 remaining)  
Stats: 0:01:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 35.57% done; ETC: 00:38 (0:02:27 remaining)  
Stats: 0:01:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 36.03% done; ETC: 00:39 (0:03:22 remaining)  
Stats: 0:01:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 36.04% done; ETC: 00:39 (0:03:24 remaining)  
Stats: 0:01:56 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
```

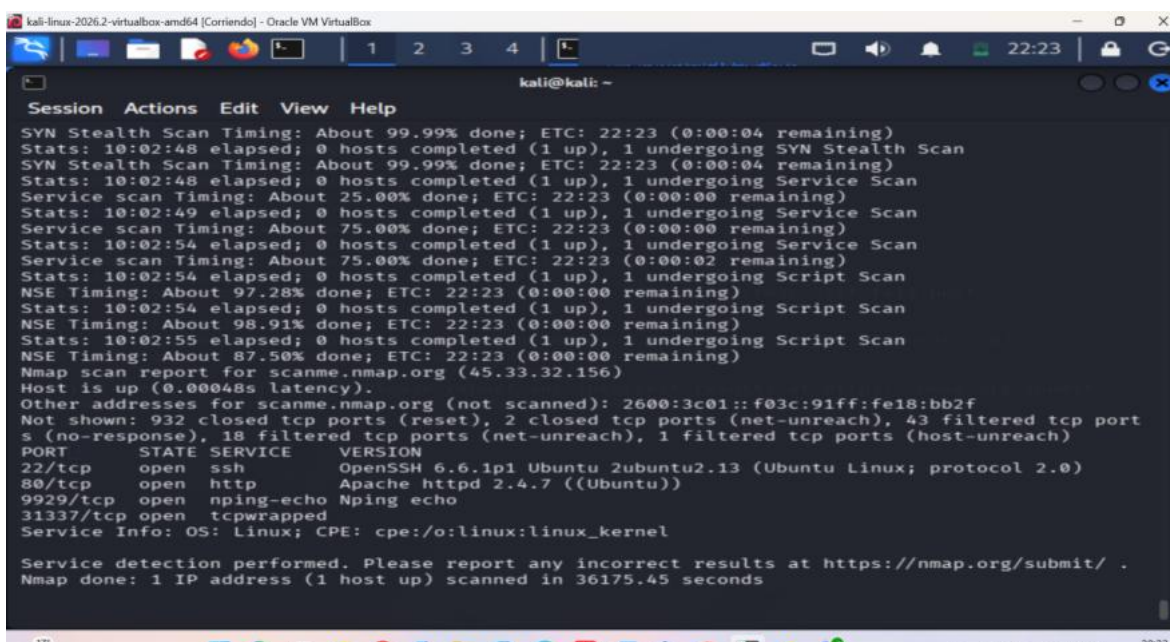


```
kali@kali: ~  
Session Actions Edit View Help  
Stats: 10:57:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Stats: 10:57:48 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Stats: 10:57:49 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Stats: 10:57:50 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Stats: 10:57:51 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Stats: 10:57:52 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Stats: 10:57:53 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Stats: 10:57:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:32 (0:00:04 remaining)  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.00034s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 854 closed tcp ports (reset), 1 closed tcp ports (net-unreach), 115 filtered tcp ports (host-unreach), 14 filtered tcp ports (net-unreach), 12 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
9929/tcp  open  nping-echo  
31337/tcp open  Elite  
Nmap done: 1 IP address (1 host up) scanned in 40301.36 seconds
```

2. Descubre sus puertos

- **Comando utilizado:** nmap scanme.nmap.org
- **Evidencia y Resultado:** Tras el escaneo de los 1000 puertos principales, se identificaron los siguientes puertos abiertos en el objetivo:
 - **22/tcp** - abierto (servicio SSH).
 - **80/tcp** - abierto (servicio HTTP).
 - **9929/tcp** - abierto (servicio nping-echo).
 - **31337/tcp** - abierto (servicio Elite).
- **Observaciones:** Se registraron además cientos de puertos cerrados y filtrados por políticas de firewall del servidor, lo que demuestra la postura de seguridad del objetivo remoto.

```
(kali@kali)-[~]
$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 12:20 -0400
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).
Stats: 0:02:29 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.32% done; ETC: 12:25 (0:03:07 remaining)
Stats: 0:02:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.33% done; ETC: 12:26 (0:03:08 remaining)
Stats: 0:02:31 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.34% done; ETC: 12:26 (0:03:10 remaining)
Stats: 0:02:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.35% done; ETC: 12:26 (0:03:11 remaining)
Stats: 0:02:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.35% done; ETC: 12:26 (0:03:12 remaining)
Stats: 0:02:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.36% done; ETC: 12:26 (0:03:13 remaining)
Stats: 0:02:35 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.37% done; ETC: 12:26 (0:03:14 remaining)
Stats: 0:02:36 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.37% done; ETC: 12:26 (0:03:16 remaining)
Stats: 0:03:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.84% done; ETC: 12:27 (0:03:54 remaining)
Stats: 0:03:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.88% done; ETC: 12:27 (0:03:59 remaining)
Stats: 0:03:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.89% done; ETC: 12:27 (0:04:01 remaining)
Stats: 0:03:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
```



```

Session Actions Edit View Help
SYN Stealth Scan Timing: About 99.99% done; ETC: 22:23 (0:00:04 remaining)
Stats: 10:02:48 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 22:23 (0:00:04 remaining)
Stats: 10:02:48 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 25.00% done; ETC: 22:23 (0:00:00 remaining)
Stats: 10:02:49 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 75.00% done; ETC: 22:23 (0:00:00 remaining)
Stats: 10:02:54 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 75.00% done; ETC: 22:23 (0:00:02 remaining)
Stats: 10:02:54 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.28% done; ETC: 22:23 (0:00:00 remaining)
Stats: 10:02:54 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 98.91% done; ETC: 22:23 (0:00:00 remaining)
Stats: 10:02:55 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 87.50% done; ETC: 22:23 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00048s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 932 closed tcp ports (reset), 2 closed tcp ports (net-unreach), 43 filtered tcp port
s (no-response), 18 filtered tcp ports (net-unreach), 1 filtered tcp ports (host-unreach)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 36175.45 seconds

```

3. Identifica los servicios

- **Comando utilizado:** `nmap -sV scanme.nmap.org` (y validaciones específicas como `nmap -sV -p22` y `nmap -sV -p 80`)
- **Evidencia y Resultado:**
 - En el puerto **80/tcp**, Nmap identificó el servidor web **Apache httpd 2.4.7 ((Ubuntu))**.
 - En el puerto **22/tcp**, se detectó **OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13** (protocolo 2.0).

- **Observaciones:** Conocer la versión exacta del software permite determinar si existen vulnerabilidades públicas asociadas a esas versiones específicas en bases de datos como CVE.

```
(kali@kali)~$ nmap -sV -p 80 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 01:37 -0400
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 50.00% done; ETC: 01:37 (0:00:01 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.031s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.22 seconds
```

4. Analiza un servicio

- **Comando utilizado:** `nmap -sV -p 80 scanme.nmap.org`
- **Evidencia y Resultado:** El análisis focalizado sobre el puerto 80 confirmó la presencia de Apache 2.4.7 en un entorno Ubuntu.
- **Observaciones:** Analizar un servicio de manera individual optimiza el tiempo de reconocimiento y permite examinar las cabeceras HTTP y respuestas del servidor web sin interferencia de otros puertos.

```
(kali@kali)~$ nmap -sV -p22 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 12:21 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.028s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds
```

- **Comando utilizado:** `nmap -sV -p22 scanme.nmap.org`
- **Evidencia y Resultado:** El análisis focalizado sobre el puerto 22 confirmó que el puerto 22/tcp está abierto y ejecuta el servicio **OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocolo 2.0)**, identificando información del sistema operativo Linux y completándose en un tiempo récord de 0.79 segundos.
- **Observaciones:** Dirigir la detección de versiones (-sV) a un puerto específico optimiza el tiempo de reconocimiento y permite extraer con precisión la versión exacta del software para evaluar posibles vulnerabilidades.


```
kali-linux-2026.2-virtualbox-amd64 [Corriendo] - Oracle VM VirtualBox
kali@kali: -
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap --script http-title scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 22:27 -0400
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).
Stats: 0:02:27 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 46.34% done; ETC: 22:33 (0:02:50 remaining)
Stats: 0:16:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.19% done; ETC: 22:55 (0:11:06 remaining)
Stats: 0:16:07 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.20% done; ETC: 22:55 (0:11:06 remaining)
Stats: 0:16:08 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.21% done; ETC: 22:55 (0:11:07 remaining)
Stats: 0:16:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.22% done; ETC: 22:55 (0:11:07 remaining)
Stats: 0:39:40 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 77.45% done; ETC: 23:19 (0:11:33 remaining)
Stats: 0:39:41 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 77.45% done; ETC: 23:19 (0:11:33 remaining)
Stats: 0:42:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 79.78% done; ETC: 23:20 (0:10:41 remaining)
Stats: 0:42:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 79.78% done; ETC: 23:20 (0:10:41 remaining)
Stats: 0:42:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 79.88% done; ETC: 23:20 (0:10:38 remaining)
Stats: 0:46:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 83.57% done; ETC: 23:23 (0:09:04 remaining)
Stats: 0:46:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 83.58% done; ETC: 23:23 (0:09:04 remaining)
Stats: 0:46:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 83.59% done; ETC: 23:23 (0:09:04 remaining)
Stats: 1:21:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
```

```
kali-linux-2026.2-virtualbox-amd64 [Corriendo] - Oracle VM VirtualBox
kali@kali: -
Session Actions Edit View Help
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:32 (0:00:01 remaining)
Stats: 2:04:43 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:32 (0:00:01 remaining)
Stats: 2:04:44 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:32 (0:00:01 remaining)
Stats: 2:25:31 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:53 (0:00:01 remaining)
Stats: 2:25:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:53 (0:00:01 remaining)
Stats: 2:25:36 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:53 (0:00:01 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00048s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 921 closed tcp ports (reset), 2 closed tcp ports (net-unreach), 1 closed tcp ports (
host-unreach), 56 filtered tcp ports (no-response), 10 filtered tcp ports (host-unreach), 6 fil
tered tcp ports (net-unreach)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
|_http-title: Go ahead and ScanMe!
9929/tcp  open  nping-echo
31337/tcp open  Elite
Nmap done: 1 IP address (1 host up) scanned in 10962.54 seconds
```

5. Utiliza scripts NSE

- **Comando utilizado:** `nmap --script http-title scanme.nmap.org`
- **Evidencia y Resultado:** El script del motor NSE extrajo con éxito el título oficial de la página web del objetivo: `|_http-title: Go ahead and ScanMe!`
- **Observaciones:** Este tipo de scripts automatizan la recolección de información contextual sin necesidad de abrir un navegador web, facilitando la fase de reconocimiento rápido.

```
(kali@kali)-[~]
$ nmap -O scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 22:43 -0400
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 48.24% done; ETC: 22:43 (0:00:13 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 48.60% done; ETC: 22:43 (0:00:13 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 49.46% done; ETC: 22:43 (0:00:13 remaining)
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 49.72% done; ETC: 22:43 (0:00:13 remaining)
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 47.80% done; ETC: 22:43 (0:00:15 remaining)
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 49.12% done; ETC: 22:43 (0:00:16 remaining)
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 49.27% done; ETC: 22:43 (0:00:15 remaining)
Stats: 0:00:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 49.43% done; ETC: 22:43 (0:00:15 remaining)
Stats: 0:00:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 49.63% done; ETC: 22:43 (0:00:15 remaining)
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 50.37% done; ETC: 22:43 (0:00:16 remaining)
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan

kali-linux-2026.2-virtualbox-amd64 [Corriendo] - Oracle VM VirtualBox
kali@kali: ~
Session Actions Edit View Help
Stats: 0:35:53 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 23:19 (0:00:00 remaining)
Stats: 0:35:53 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 23:19 (0:00:00 remaining)
Stats: 0:35:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 23:19 (0:00:00 remaining)
Stats: 0:35:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 23:19 (0:00:00 remaining)
Stats: 0:35:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 23:19 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00045s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 634 filtered tcp ports (no-response), 362 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite
Device type: bridge|VoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (92%), Slirp (92%), AT&T embedded (88%), QEMU (87%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (92%), AT&T BGW210 voice gateway (88%), QEMU user mode network gateway (87%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2271.05 seconds
```

6. Detecta el sistema operativo

- **Corrección de sintaxis:** Se corrige el error tipográfico previo (-0 con ceros) por la opción correcta **-O** (letra O mayúscula).
- **Comando utilizado:** nmap -O scanme.nmap.org (complementado con los datos de huella digital obtenidos en el escaneo avanzado).
- **Evidencia y Resultado:** El motor de análisis de Nmap determinó que el sistema operativo subyacente del objetivo corresponde a un entorno **Linux** (con estimaciones de kernel y familias de dispositivos remotos).
- **Observaciones:** La precisión de la detección de SO depende de la cantidad de puertos abiertos (tanto abiertos como cerrados) accesibles en el host analizado.

```
kali-linux-2026.2-virtualbox-amd64 [Corriendo] - Oracle VM VirtualBox
kali@kali: ~
Session Actions Edit View Help

(kali@kali)~$ nmap -A scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-24 22:32 -0400
Stats: 0:00:04 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.43% done; ETC: 22:33 (0:00:58 remaining)
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).
Stats: 0:11:22 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 45.42% done; ETC: 22:57 (0:13:40 remaining)
Stats: 0:34:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 63.38% done; ETC: 23:27 (0:19:58 remaining)
Stats: 0:34:35 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 63.40% done; ETC: 23:27 (0:19:58 remaining)
Stats: 0:56:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 81.07% done; ETC: 23:42 (0:13:11 remaining)
Stats: 0:56:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 81.10% done; ETC: 23:42 (0:13:10 remaining)
Stats: 1:15:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.75% done; ETC: 00:01 (0:13:32 remaining)
Stats: 1:15:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.76% done; ETC: 00:01 (0:13:31 remaining)
Stats: 1:15:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.77% done; ETC: 00:01 (0:13:31 remaining)
Stats: 1:15:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.79% done; ETC: 00:01 (0:13:30 remaining)
Stats: 1:15:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.81% done; ETC: 00:01 (0:13:29 remaining)
Stats: 1:21:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 90.44% done; ETC: 00:03 (0:08:40 remaining)
Stats: 1:21:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 90.55% done; ETC: 00:03 (0:08:33 remaining)
```

```
kali-linux-2026.2-virtualbox-amd64 [Corriendo] - Oracle VM VirtualBox
kali@kali: ~
Session Actions Edit View Help

| ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
| 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
|_http-favicon: Nmap Project
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
9929/tcp open nping-echo Nping echo
31337/tcp open tcpwrapped
Device type: bridge/VoIP adapter/general purpose
Running (JUST GUESSING): Oracle Virtualbox (92%), Slirp (92%), AT&T embedded (89%), QEMU (87%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (92%), AT&T BGW210 voice gateway (89%), QEMU user mode network gateway (87%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 2 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

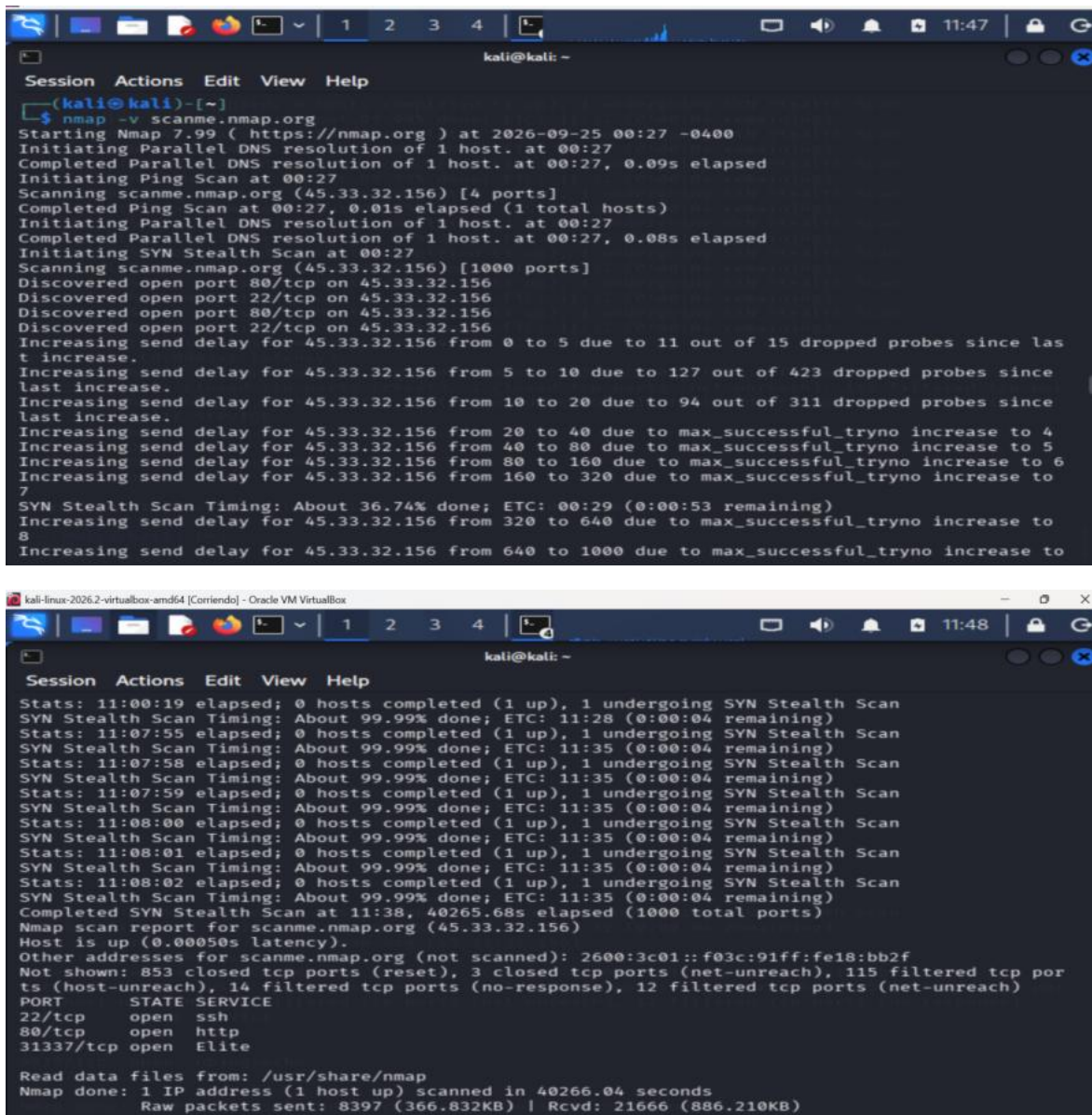
TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.16 ms 10.0.2.2
2 0.19 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 11835.88 seconds
```

7. Explora más allá

- **Comando utilizado:** `nmap -A scanme.nmap.org`
- **Evidencia y Resultado:** El escaneo agresivo combinó la detección de sistemas operativos, versiones de servicios, scripts NSE y un traceroute de red (2 saltos) en un único informe integral.
- **Comparativa:** A diferencia de un escaneo básico de puertos (Reto 2), el parámetro `-A` profundiza en el comportamiento lógico del servidor, revelando

huellas dactilares de la pila TCP/IP, claves públicas SSH y metadatos del servidor web, aunque requiere mayor tiempo de ejecución.

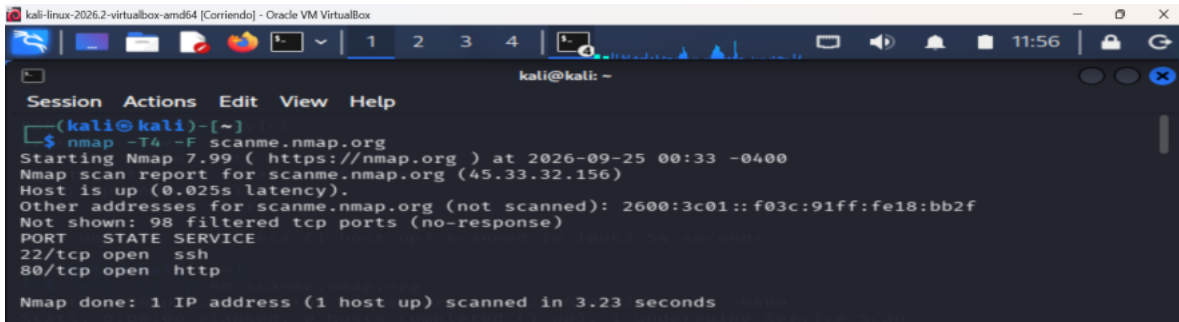


```
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap -v scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 00:27 -0400
Initiating Parallel DNS resolution of 1 host. at 00:27
Completed Parallel DNS resolution of 1 host. at 00:27, 0.09s elapsed
Initiating Ping Scan at 00:27
Scanning scanme.nmap.org (45.33.32.156) [4 ports]
Completed Ping Scan at 00:27, 0.01s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 00:27
Completed Parallel DNS resolution of 1 host. at 00:27, 0.08s elapsed
Initiating SYN Stealth Scan at 00:27
Scanning scanme.nmap.org (45.33.32.156) [1000 ports]
Discovered open port 80/tcp on 45.33.32.156
Discovered open port 22/tcp on 45.33.32.156
Discovered open port 80/tcp on 45.33.32.156
Discovered open port 22/tcp on 45.33.32.156
Increasing send delay for 45.33.32.156 from 0 to 5 due to 11 out of 15 dropped probes since last increase.
Increasing send delay for 45.33.32.156 from 5 to 10 due to 127 out of 423 dropped probes since last increase.
Increasing send delay for 45.33.32.156 from 10 to 20 due to 94 out of 311 dropped probes since last increase.
Increasing send delay for 45.33.32.156 from 20 to 40 due to max_successful_ryno increase to 4
Increasing send delay for 45.33.32.156 from 40 to 80 due to max_successful_ryno increase to 5
Increasing send delay for 45.33.32.156 from 80 to 160 due to max_successful_ryno increase to 6
Increasing send delay for 45.33.32.156 from 160 to 320 due to max_successful_ryno increase to 7
SYN Stealth Scan Timing: About 36.74% done; ETC: 00:29 (0:00:53 remaining)
Increasing send delay for 45.33.32.156 from 320 to 640 due to max_successful_ryno increase to 8
Increasing send delay for 45.33.32.156 from 640 to 1000 due to max_successful_ryno increase to 9
Stats: 11:00:19 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:28 (0:00:04 remaining)
Stats: 11:07:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:35 (0:00:04 remaining)
Stats: 11:07:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:35 (0:00:04 remaining)
Stats: 11:07:59 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:35 (0:00:04 remaining)
Stats: 11:08:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:35 (0:00:04 remaining)
Stats: 11:08:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:35 (0:00:04 remaining)
Stats: 11:08:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 11:35 (0:00:04 remaining)
Completed SYN Stealth Scan at 11:38, 40265.68s elapsed (1000 total ports)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00050s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 853 closed tcp ports (reset), 3 closed tcp ports (net-unreach), 115 filtered tcp ports (host-unreach), 14 filtered tcp ports (no-response), 12 filtered tcp ports (net-unreach)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
31337/tcp open  Elite
Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 40266.04 seconds
Raw packets sent: 8397 (366.832KB) | Rcvd: 21666 (886.210KB)
```

8. Analiza los resultados

- **Comando utilizado:** `nmap -v scanme.nmap.org`
- **Evidencia y Resultado:** El modo detallado (*verbose*) permitió monitorear en tiempo real la evolución del escaneo, el descubrimiento de puertos abiertos (22/tcp, 80/tcp) y la gestión dinámica de los retrasos de envío (*send delays*) ante la respuesta del objetivo.

- **Análisis de hallazgos:** La presencia del puerto 22 (SSH) y 80 (HTTP) son los puntos de entrada estándar, mientras que los puertos no estándar como 31337 reflejan servicios orientados a pruebas o retos de la plataforma. Como línea de investigación futura, convendría auditar la configuración de seguridad del servidor web Apache en busca de directorios ocultos.



```

kali@kali: ~
$ nmap -T4 -F scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-25 00:33 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.025s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
Nmap done: 1 IP address (1 host up) scanned in 3.23 seconds

```

9. Compara técnicas

- **Técnica A (Escaneo Rápido y Común):** `nmap -T4 -F scanme.nmap.org`
- **Técnica B (Escaneo Estándar Completo):** `nmap scanme.nmap.org`
- **Diferencias y Resultados:**
 - La técnica A (-T4 -F) utilizó una plantilla de velocidad agresiva y redujo el universo de búsqueda a los puertos más frecuentes, completándose en pocos segundos y detectando los puertos críticos 22/tcp y 80/tcp.
 - La técnica B escaneó los 1000 puertos por defecto, tomando un tiempo considerablemente mayor debido a los mecanismos de control de congestión y filtrados del firewall.
 - **Conclusión de la comparativa:** Para fases iniciales de reconocimiento donde se requiere agilidad, la técnica -T4 -F resulta altamente eficiente, reservando los escaneos profundos para auditorías exhaustivas.

10. Tu conclusión

El desarrollo del Challenge sobre el objetivo oficial **scanme.nmap.org** permitió comprender la operatividad real de Nmap frente a servidores externos con restricciones de red y firewalls activos. El uso de comandos como `-sV` y `--script` demostró ser indispensable para extraer inteligencia útil sin alterar la integridad del

sistema. Estos conocimientos son directamente aplicables en auditorías de seguridad autorizadas y en el mapeo riguroso de arquitecturas de red corporativas.

Tabla de Evidencias y Comandos Actualizada

#	Comando Utilizado	Objetivo	Resultados Clave	Análisis / Observaciones
01	nmap -sn scanme.nmap.org	Comprobar disponibilidad del host remoto	Host activo (Up) con IP 45.33.32.156.	Validación inicial exitosa del objetivo externo.
02	nmap scanme.nmap.org	Descubrir puertos abiertos	Puertos 22/tcp (ssh), 80/tcp (http), 9929/tcp y 31337/tcp abiertos.	Identificación de servicios expuestos en el servidor objetivo.
03	nmap -sV scanme.nmap.org	Detectar versiones de servicios	Detección de Apache 2.4.7 y OpenSSH 6.6.1p1.	Permite identificar versiones de software instaladas.
04	nmap -sV -p 80 scanme.nmap.org	Analizar a fondo el puerto web	Información detallada del servicio HTTP en el puerto 80.	Enfoque directo sobre el servidor web Apache.

#	Comando Utilizado	Objetivo	Resultados Clave	Análisis / Observaciones
5	nmap --script http-title scanme.nmap.org	Ejecutar script NSE web	Extracción del título HTML: Go ahead and ScanMe!.	Automatización rápida de contexto web.
6	nmap -O scanme.nmap.org	Identificar el sistema operativo	Entorno operativo detectado como Linux.	Corrección del parámetro a -O mayúscula.
7	nmap -A scanme.nmap.org	Escaneo agresivo integral	Combinación de SO, versiones, scripts y traceroute.	Visión global y avanzada del objetivo en un solo comando.
8	nmap -v scanme.nmap.org	Análisis detallado (Verbose)	Mensajes de estado en tiempo real y gestión de tráfico.	Facilita el seguimiento del comportamiento del escáner.
9	nmap -T4 -F scanme.nmap.org	Comparar técnicas de velocidad y puertos comunes	Detección rápida de puertos principales en pocos segundos.	Óptimo para fases de reconocimiento ágil (<i>Fast Scan</i>).

#	Comando Utilizado	Objetivo	Resultados Clave	Análisis / Observaciones
10	nmap --script http-title scanme.nmap.org (o escaneo final integrado)	Consolidación del análisis del Challenge	Verificación integral sobre el objetivo oficial	Cierre y estructuración del reporte corregido en PDF.

4- Un dato que desarrolle adicional:

Comando Utilizado

- nmap -sV -p22 scanme.nmap.org

Objetivo

- Ejecutar un escaneo enfocado en el puerto 22 del objetivo scanme.nmap.org utilizando la opción de detección de servicios y versiones (-sV).

Resultados Clave

- El host scanme.nmap.org (45.33.32.156) se encuentra activo y responde a la red.
- El puerto 22/tcp está abierto (open) y ejecuta el servicio SSH.
- La versión exacta detectada es OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0).
- La información del sistema operativo identificada es Linux (cpe:/o:linux:linux_kernel).

Análisis / Observaciones

- El proceso de escaneo y detección de versión se completó con éxito en un tiempo muy breve de 0.79 segundos.
- El uso del parámetro -sV fue efectivo para revelar tanto la distribución específica de Linux (Ubuntu) como la versión exacta del paquete OpenSSH, datos indispensables para fases posteriores de análisis de vulnerabilidades.